

Problem

Use the Laplace transform to find $i(t)$ for $t > 0$ if

$$\frac{d^2 i}{dt^2} + 3 \frac{di}{dt} + 2i + \delta(t) = 0,$$

$$i(0) = 0, \quad i'(0) = 3$$

Step-by-step solution

Step 1 of 1

$$\frac{d^2 i}{dt^2} + 3 \frac{di}{dt} + 2i + \delta(t) = 0$$

$$j(0) = 0, \quad j'(0) = 3.$$

Taking $L.T$

$$S^2 I(S) - S j(0) - j'(0) + 3[S I(S) - j(0)] + 2 I(S) + 1 = 0$$

$$\Rightarrow I(S)[S^2 + 3S + 2] - S(0) - 3 + 3(0) + 1 = 0$$

$$I(S) = \frac{2}{S^2 + 3S + 2} = \frac{2}{(S+1)(S+2)}$$

$$= 2 \left[\frac{1}{S+1} - \frac{1}{S+2} \right]$$

$$j(t) = \boxed{2(e^{-t} - e^{-2t})u(t)}.$$